SOFTWARE ENGINEERING

Scrum Roles, Product Owner and Product Backlog

A detailed 90-minute teaching lecture for BS-level students

Lecture focus
Scrum accountabilities (often informally called roles), the Product Owner's responsibilities, and how the Product Backlog is created, ordered, refined, and used during product development.

Suggested duration: 90 minutes
Audience: BS Software Engineering / computing students
Teaching approach: Explanation, practical example, short classroom activity, and review questions.

Prepared as instructor-friendly notes with definitions, examples, board prompts, and a short assessment.

1. Lecture Plan (90 Minutes)

Time	Section	Teaching method / outcome
0–5 min	Opening and learning outcomes	Connect Scrum to iterative software delivery; introduce key terms.
5–15 min	Scrum and its purpose	Explain Scrum as a lightweight framework for complex work.
15–35 min	Scrum Team and accountabilities	Compare Product Owner, Scrum Master, and Developers.
35–50 min	Product Owner in detail	Discuss value, decisions, stakeholder input, and common misconceptions.
50–68 min	Product Backlog in detail	Explain backlog items, ordering, refinement, transparency, and changes.
68–80 min	Worked example and class activity	Build and order a small backlog for a university portal.
80–87 min	Quick quiz and discussion	Check understanding through short questions.
87–90 min	Summary and exit ticket	Students state one responsibility and one backlog principle.

2. Learning Outcomes

By the end of this lecture, students should be able to:

- Explain what Scrum is and why teams use it when developing products in complex environments.
- Identify the three Scrum accountabilities: Product Owner, Scrum Master, and Developers.
- Describe the Product Owner's accountability for maximizing product value and managing the Product Backlog.
- Define a Product Backlog Item (PBI) and distinguish a backlog from a fixed, unchangeable task list.
- Explain how backlog ordering, refinement, transparency, and the Product Goal support delivery.
- Create and prioritize a small sample Product Backlog using value, risk, urgency, and dependencies.

3. Opening: What Is Scrum? (5–15 Minutes)

Suggested opening words: "Imagine that a university wants a student portal. Students need to view attendance, teachers need to upload marks, and administrators need reports. The team cannot know every detail perfectly on day one. Scrum helps the team deliver useful parts in short cycles, receive feedback, and adapt as it learns."

Scrum is a lightweight framework that helps people and teams generate value through adaptive solutions for complex problems. It is based on empiricism and lean thinking. Empiricism means making decisions using what is observed and learned; lean thinking emphasizes reducing waste and focusing on value.

Key ideas to explain

- **Iterative:** the team revisits and improves the product through repeated cycles.
- **Incremental:** the product grows by adding usable pieces of functionality.
- **Empirical:** decisions are informed by evidence, inspection, and experience rather than assumptions alone.
- **Timeboxed work:** Scrum uses Sprints of a fixed duration, one month or less.

> A Sprint is not simply a deadline for writing code. It is a short period in which the Scrum Team works toward a Sprint Goal and aims to produce a usable, valuable Increment that meets the Definition of Done.

Three pillars of empiricism

Pillar	Meaning	Simple example
Transparency	Important work and its status are visible and understandable.	The backlog clearly shows what is being considered and why.
Inspection	People regularly examine progress and results.	The team reviews a working portal feature and checks quality.
Adaptation	The team changes its approach when evidence shows a problem or opportunity.	Student feedback causes the team to improve the attendance screen.

4. The Scrum Team and Its Accountabilities (15–35 Minutes)

The Scrum Team consists of one Scrum Master, one Product Owner, and Developers. These are accountabilities, not job titles that must correspond to exactly three individual employees. One person may hold a job title such as developer or analyst, but within Scrum the accountability is defined by the work and responsibilities. The Scrum Team is designed to be cross-functional and self-managing.

4.1 Product Owner

The Product Owner is accountable for maximizing the value of the product resulting from the work of the Scrum Team. The Product Owner is also accountable for effective Product Backlog management.

- Developing and communicating the Product Goal.
- Creating and clearly communicating Product Backlog items.
- Ordering Product Backlog items.
- Ensuring that the Product Backlog is transparent, visible, and understood.
- May do the work personally or delegate it, but remains accountable.

Example: For a university portal, the Product Owner decides that reliable student login and attendance viewing should be addressed before a decorative theme because the former provides more immediate value.

4.2 Scrum Master

The Scrum Master is accountable for establishing Scrum as defined in the Scrum Guide and for the Scrum Team's effectiveness. The Scrum Master helps people understand Scrum theory and practice.

- Coaches team members in self-management and cross-functionality.
- Helps the team focus on creating valuable, high-quality Increments.
- Helps remove impediments to progress, or helps the right people address them.
- Ensures Scrum events occur and are productive, within the Scrum framework.
- Supports the Product Owner with effective Product Goal and Product Backlog management practices.

The Scrum Master is not simply a boss, project secretary, or person who assigns every task. The role is one of leadership and service.

4.3 Developers

Developers are the people in the Scrum Team who are committed to creating any aspect of a usable Increment each Sprint. "Developers" does not mean only programmers; it can include the skills needed to produce the product.

- Create a plan for the Sprint, the Sprint Backlog.
- Instill quality by adhering to the Definition of Done.
- Adapt their plan each day toward the Sprint Goal.
- Hold each other accountable as professionals.

For a student portal, Developers might include programmers, testers, UX designers, data specialists, and others who collectively have the skills needed to create a usable increment.

Accountability	Primary focus	Example question
Product Owner	Product value and Product Backlog	“What is the most valuable thing to work on next?”
Scrum Master	Scrum effectiveness and enabling the team	“How can we improve our way of working?”
Developers	Creating a usable Increment and planning the work	“How can we deliver this Sprint Goal with quality?”

Important distinction: shared work, clear accountability

The three accountabilities collaborate closely, but they are not interchangeable. The Product Owner is accountable for product value and backlog management. The Scrum Master is accountable for Scrum’s effectiveness. Developers decide how to turn selected work into a usable Increment. No one outside the Scrum Team dictates how Product Backlog items are turned into Increments.

5. The Product Owner in Detail (35–50 Minutes)

5.1 The Product Owner’s purpose

A Product Owner helps the Scrum Team focus on the product outcomes that matter most. The role connects user needs, organizational goals, and the team’s development work. The Product Owner makes ordering decisions and communicates the product direction.

5.2 Product Goal

The Product Goal describes a future state of the product and provides a target for the Scrum Team to plan against. It gives the Product Backlog a longer-term objective. The Scrum Team should fulfill, or abandon, one Product Goal before taking on the next.

Example Product Goal: “Provide students and faculty with a reliable portal for checking attendance and publishing course marks.” This is a product outcome, not a list of technical tasks.

5.3 Working with stakeholders

Stakeholders may include students, teachers, department heads, administrators, IT support, and people responsible for privacy or security. The Product Owner gathers and evaluates their needs, but does not automatically accept every request. Requests can conflict, so the Product Owner considers user value, strategic goals, risk, cost, dependencies, and evidence.

- **Listen:** understand the problem and who experiences it.
- **Clarify:** ask what success would look like and what constraints apply.
- **Compare:** assess relative value, risk, urgency, effort, and dependencies.
- **Decide and communicate:** make the ordering clear and explain the reasoning.

5.4 What the Product Owner is not

- Not necessarily the person who writes all the code or personally performs every backlog task.
- Not simply a messenger who accepts every stakeholder request.
- Not the sole person allowed to talk to customers or users; collaboration is encouraged.
- Not the manager who assigns detailed technical tasks to Developers.
- Not guaranteed to be the project manager; Product Owner is a distinct Scrum accountability.

5.5 A practical decision

Suppose stakeholders request three items: (A) students can view attendance, (B) a profile page can use custom colors, and (C) login security is improved. The Product Owner should ask about user impact, security exposure, dependencies, and the product goal. Security and attendance may reasonably rank above visual customization, but the correct order depends on the context and evidence—not on a universal fixed rule.

6. Product Backlog in Detail (50–68 Minutes)

6.1 Definition

The Product Backlog is an emergent, ordered list of what is needed to improve the product. It is the single source of work undertaken by the Scrum Team. It is not necessarily a complete specification written once at the beginning; it changes as the team learns more about users, technology, business needs, and the product.

6.2 Product Backlog Items (PBIs)

A Product Backlog Item is an individual piece of work or product improvement represented in the backlog. Items may describe user-facing functionality, defects, technical improvements, research, compliance work, or other needs. Scrum does not require one particular template for every item.

Example PBI: “As a student, I want to view my attendance by course so that I can identify classes where I need to follow up.” A user story is a common way to phrase an item, but it is not the only valid format.

6.3 Common information in a backlog item

- **Description:** what need, opportunity, defect, or improvement is being addressed.
- **Value or purpose:** why the item matters and for whom.
- **Acceptance examples/criteria:** examples that help clarify expected behavior; these are useful practices, not a mandatory Scrum template.
- **Estimate:** a team’s relative estimate of effort or complexity, if the team uses estimation.
- **Dependencies and risks:** other work, decisions, or uncertainties that may affect delivery.

The Product Backlog itself must be transparent. The precise fields, tool, and estimation technique are up to the team, provided they support understanding and effective decisions.

6.4 Ordering the Product Backlog

The Product Owner orders Product Backlog items to optimize value and progress toward the Product Goal. Ordering is more thoughtful than simply sorting items by who asked first. Useful considerations include:

- Value and impact for users or the organization.
- Risk reduction, security, privacy, and regulatory obligations.
- Urgency and time-sensitive opportunities.
- Dependencies: whether one item enables another.
- Learning: whether an item reduces a major uncertainty.
- Cost and effort, considered alongside value rather than in isolation.

> **Priority versus order:** teams often say “priority,” but the Scrum Guide uses “ordered.” An ordered backlog communicates the current sequence of importance while allowing the order to change as evidence changes.

6.5 Product Backlog refinement

Product Backlog refinement is the ongoing activity of breaking down and further defining Product Backlog items into smaller, more precise items. Details such as description, order, and size are added or updated. Developers who will do the work participate in sizing. Refinement is not a formal Scrum event and there is no fixed amount of time prescribed for it.

For example, “Improve attendance reporting” may be too broad. Refinement may split it into: view attendance by course, filter by date range, export a report, and show an understandable absence summary. The team can then discuss examples, dependencies, risks, and relative size.

6.6 Product Backlog and Sprint Backlog are different

Product Backlog	Sprint Backlog
The ordered, evolving list of what may improve the product.	The Sprint Goal, selected Product Backlog items for the Sprint, and actionable plan for delivering them.
Product Owner is accountable for effective management.	Created and adapted by Developers; it is a plan by and for them.
Can change as the product and context are learned.	Updated throughout the Sprint as more is learned, while protecting the Sprint Goal.

6.7 Transparency and change

The backlog should be visible and understandable to the people who need it. It is not a contract promising that every item will be built. New information may change what is most valuable. The Product Owner can reorder the Product Backlog as needed. During a Sprint, scope may be clarified and renegotiated with the Product Owner as more is learned, but changes should not endanger the Sprint Goal or reduce quality.

A backlog that is merely a long list of vague requests is not enough. Effective management means making the product direction clear, keeping items understandable, and ordering them so the team can make good decisions.

7. Worked Example: University Student Portal

Use this example on the board or projector. Ask students to imagine they are the Scrum Team supporting a university that currently uses paper forms and scattered spreadsheets.

Possible Product Goal

“Enable students and faculty to manage essential academic information through a reliable online portal.”

Order (example)	Product Backlog Item	Why it may be ordered here
1	Secure sign-in and account access	Access and security are foundational; the actual order depends on risk and current systems.
2	View attendance by course	Direct student need; supports timely action.
3	Faculty can publish course marks	High academic value, subject to approval and privacy rules.
4	Show a clear attendance summary	Helps students understand attendance without manual calculation.
5	Export attendance report	Useful for staff; may depend on attendance data being available.
6	Customize profile colors	Potentially nice to have, but likely less valuable than core functions.

Teaching note: stress that this is an illustrative order, not the one correct order for every university. If a legal deadline, urgent security weakness, or critical dependency changes, the Product Owner may need to reorder the items.

Short classroom activity (8–10 minutes)

Instructions: Put students in pairs or small groups. Ask each group to choose the top three items for a first release and defend the ordering in two sentences.

- View attendance by course.

- Secure sign-in.
- Upload a profile photo.
- Faculty can publish marks.
- Download a fee challan.
- Dark mode.
- Report an incorrect attendance entry.

Discussion prompts: What user problem is most important? Is there a security or compliance concern? Does one item depend on another? What information would you want before deciding? Could two items be valuable but in a different order for a different university?

> **Expected learning:** ordering is a reasoned, context-sensitive decision; it is not a popularity contest or simply the shortest task first.

8. Common Misunderstandings to Correct

1. **“The Product Owner is the boss of Developers.”** Not in the sense of dictating how they perform technical work. The Product Owner is accountable for product value and Product Backlog management.
2. **“The Scrum Master assigns tasks.”** The Scrum Master helps establish Scrum and improve effectiveness. Developers manage their work and plan how to achieve the Sprint Goal.
3. **“Developers means only programmers.”** Developers are the people with the skills needed to create a usable Increment; the group may include several specialties.
4. **“The Product Backlog must be complete before development starts.”** It evolves as knowledge and context change.
5. **“The highest-priority item can never change.”** The Product Owner can reorder items when evidence, risks, or goals change.
6. **“Every backlog item must be a user story with story points.”** These are possible techniques, not universal Scrum requirements.
7. **“Refinement is a formal Scrum event.”** Refinement is an ongoing activity, not one of the five formal Scrum events.

9. Quick Assessment (80–87 Minutes)

Ask students to answer individually first, then review answers aloud.

1. Name the three Scrum accountabilities.
2. What is the Product Owner accountable for?
3. What is the purpose of the Product Goal?
4. Define a Product Backlog Item and give one example.
5. Give two factors—other than who asked first—that can influence backlog ordering.
6. What is the difference between the Product Backlog and Sprint Backlog?
7. Is Product Backlog refinement a formal Scrum event? Explain briefly.
8. A stakeholder asks for a decorative feature, but the team has identified a serious security concern. What should influence the Product Owner’s decision?

Answer Key

1. Product Owner, Scrum Master, and Developers.
2. Maximizing product value and effective Product Backlog management.
3. It describes a future state of the product and provides a target for the Scrum Team to plan against.

4. A piece of work or improvement in the backlog; for example, “students can view attendance by course.”
5. Examples: user value, risk, urgency, dependencies, cost/effort, and learning opportunities.
6. The Product Backlog is the evolving ordered list of product work; the Sprint Backlog is the Sprint Goal, selected items, and the Developers’ plan.
7. No. It is an ongoing activity used to break down and clarify items, update their order, and consider their size.
8. User impact, severity of risk, product goal, urgency, dependencies, and relevant security/privacy obligations; the decision should be explained and reflected in backlog ordering.

10. Closing Summary (87–90 Minutes)

Close the lecture with these key messages:

- Scrum has three accountabilities: Product Owner, Scrum Master, and Developers.
- The Product Owner focuses on product value and is accountable for effective Product Backlog management.
- The Product Backlog is ordered, evolving, and transparent—not a frozen list of promises.
- Refinement improves the clarity and size of backlog items as the team learns.
- Good ordering balances value, risk, urgency, dependencies, and the Product Goal.

Exit ticket: Ask every student to write one sentence explaining the Product Owner’s accountability and one factor that could cause a Product Backlog item to move up or down the order.

11. Instructor Reference Notes

This lecture is aligned with the core terminology and framework described in the official Scrum Guide (2020 edition): the Scrum Team and its accountabilities, Product Goal, Product Backlog, Product Backlog items, ordering, and refinement. The guide is concise by design; this lecture expands those concepts with teaching examples and classroom exercises. Local course outlines may differ in sequence or emphasis.

Recommended source: Schwaber, Ken and Jeff Sutherland, *The Scrum Guide* (November 2020), available at scrumguides.org. Use the official guide as the primary reference if students need the formal definitions.

Preparation tip: Before class, draw three columns on the board—Product Owner, Scrum Master, Developers—and ask students to place sample responsibilities under the correct accountability. Then draw a short ordered backlog and show how new evidence can change the order.